source

File: source.kicad_sch

harmonic_1

File: harmonic_1.kicad_sch

harmonic_3

File: harmonic_3.kicad_sch

harmonic_5

File: harmonic_5.kicad_sch

harmonic_7

File: harmonic_7.kicad_sch

power_sum

File: power_sum.kicad_sch

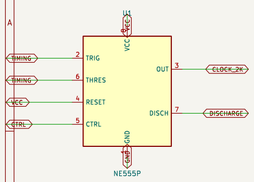
NE555 2 kHz -> CD4013 /2 -> 10 Vpp SQ -> 1/3/5/7 kHz BPF -> phase / amplitude -> sum
External regulated +12 V / GND / −12 V. High impedance output measurement.

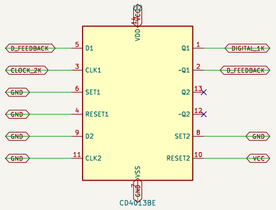
Sheet: /
File: waveform.kicad_sch

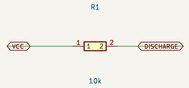
**Title: 1 kHz Fourier waveform synthesis**

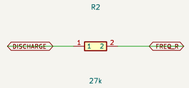
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-29 | **Rev: A** |
| KiCad E.D.A. 9.0.2 | | Id: 1/7 |

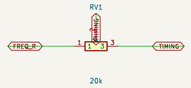
All connections use named global nets. VCC = +12V; VEE = −12V; GND = 0V. See design notes for topology and calibration.



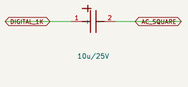



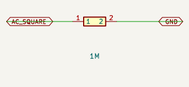



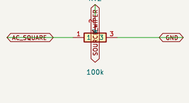


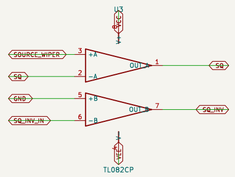


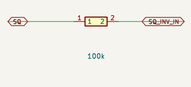


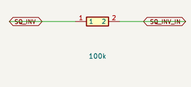













**NE555 + TL082 / +/−12 V**

Sheet: /source/
File: source.kicad_sch

***Title: Waveform synthesis / source***

| Size: A2 | Date: 2026-10-29 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 9.0.2 | | Id: 2/7 |

All connections use named global nets. VCC=+12V; VEE=−12V; GND=0V. See design notes for topology and calibration.

R6
SQ
1 2
H1_NODE1
220k

R7
H1_NODE1
1 2
H1_TUNE1
430

RV3
+12V
H1_TUNE1
1 3
GND
200

R8
H1_BPF1
1 2
H1_INV1
470k

C4
H1_NODE1
1 2
H1_BPF1
10n

C5
H1_NODE1
1 2
H1_INV1
10n

R9
H1_BPF1
1 2
H1_NODE2
220k

R10
H1_NODE2
1 2
H1_TUNE2
430

RV4
+12V
H1_TUNE2
1 3
GND
200

R11
H1_BPF2
1 2
H1_INV2
470k

C6
H1_NODE2
1 2
H1_BPF2
10n

C7
H1_NODE2
1 2
H1_INV2
10n

U4
+12V
GND
H1_INV1
GND
H1_INV2
+A
−A
+B
−B
OUT
OUT
1
2
5
6
7
H1_BPF1
H1_BPF2
-12V
TL082CP

R12
H1_BPF2
1 2
H1_APINV1
10k

R13
H1_AP1
1 2
H1_APINV1
10k

R14
H1_BPF2
1 2
H1_PHR1
100

RV5
+12V
H1_PHR1
1 3
H1_LP1
200k

C8
H1_LP1
1 2
GND
10n

R15
H1_AP1
1 2
H1_APINV2
10k

R16
H1_AP2
1 2
H1_APINV2
10k

R17
H1_AP1
1 2
H1_PHR2
100

RV6
+12V
H1_PHR2
1 3
H1_LP2
200k

C9
H1_LP2
1 2
GND
10n

U5
+12V
H1_LP1
H1_APINV1
H1_LP2
H1_APINV2
+A
−A
+B
−B
OUT
OUT
1
2
5
6
7
H1_AP1
H1_AP2
-12V
TL082CP

R18
H1_AP2
1 2
H1_POLINV
10k

R19
H1_NEG
1 2
H1_POLINV
10k

J1
H1_AP2
H1_SELECT
H1_NEG
1
2
3
PHASE_POLARITY

RV7
+12V
H1_SELECT
1 3
GND
10k

U6
+12V
GND
H1_POLINV
H1_AMPLITUDE
H1_OUT
+A
−A
+B
−B
OUT
OUT
1
2
5
6
7
H1_NEG
H1_OUT
-12V
TL082CP

TP4
H1_BPF2

TP5
H1_AP2

TP6
H1_OUT

NE555 + TL082 / +/−12 V
Sheet: /harmonic_1/
File: harmonic_1.kicad_sch
Title: Waveform synthesis / harmonic_1
Size: A2
Date: 2026-10-29
Rev: A
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Id: 3/7

All connections use named global nets. VCC=+12V; VEE=−12V; GND=0V. See design notes for topology and calibration.

R20
SQ
1 2
H3_NODE1
220k

R21
H3_NODE1
1 2
H3_TUNE1
430

RV8
VCC
H3_TUNE1
1 3
GND
200

R22
H3_BPF1
1 2
H3_INV1
470k

C10
H3_NODE1
1 2
H3_BPF1
3.3n

C11
H3_NODE1
1 2
H3_INV1
3.3n

R23
H3_BPF1
1 2
H3_NODE2
220k

R24
H3_NODE2
1 2
H3_TUNE2
430

RV9
VCC
H3_TUNE2
1 3
GND
200

R25
H3_BPF2
1 2
H3_INV2
470k

C12
H3_NODE2
1 2
H3_BPF2
3.3n

C13
H3_NODE2
1 2
H3_INV2
3.3n

U7
VCC
GND
3
+A
H3_INV1
2
−A
OUT
1
H3_BPF1
GND
5
+B
H3_INV2
6
−B
OUT
7
H3_BPF2
VEE
TL082CP

R26
H3_BPF2
1 2
H3_APINV1
10k

R27
H3_AP1
1 2
H3_APINV1
10k

R28
H3_BPF2
1 2
H3_PHR1
100

RV10
VCC
H3_PHR1
1 3
H3_LP1
200k

C14
H3_LP1
1 2
GND
3.3n

R29
H3_AP1
1 2
H3_APINV2
10k

R30
H3_AP2
1 2
H3_APINV2
10k

R31
H3_AP1
1 2
H3_PHR2
100

RV11
VCC
H3_PHR2
1 3
H3_LP2
200k

C15
H3_LP2
1 2
GND
3.3n

U8
VCC
H3_LP1
3
+A
H3_APINV1
2
−A
OUT
1
H3_AP1
H3_LP2
5
+B
H3_APINV2
6
−B
OUT
7
H3_AP2
VEE
TL082CP

R32
H3_AP2
1 2
H3_POLINV
10k

R33
H3_NEG
1 2
H3_POLINV
10k

J2
H3_AP2
H3_SELECT
H3_NEG
1
2
3
PHASE_POLARITY

RV12
VCC
H3_SELECT
1 3
GND
10k

U9
VCC
GND
3
+A
H3_POLINV
2
−A
OUT
1
H3_NEG
H3_AMPLITUDE
5
+B
H3_OUT
6
−B
OUT
7
H3_OUT
VEE
TL082CP

TP7
H3_BPF2

TP8
H3_AP2

TP9
H3_OUT

NE555 + TL082 / +/−12 V
Sheet: /harmonic_3/
File: harmonic_3.kicad_sch
Title: Waveform synthesis / harmonic_3
Size: A2
Date: 2026-10-29
Rev: A
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Id: 4/7

All connections use named global nets. VCC=+12V; VEE=−12V; GND=0V. See design notes for topology and calibration.

R34
SQ
1 2
H5_NODE1
220k

R35
H5_NODE1
1 2
H5_TUNE1
430

RV13
+12V
H5_TUNE1
1 3
GND
200

R36
H5_BPF1
1 2
H5_INV1
470k

C16
H5_NODE1
1 2
H5_BPF1
2n

C17
H5_NODE1
1 2
H5_INV1
2n

R37
H5_BPF1
1 2
H5_NODE2
220k

R38
H5_NODE2
1 2
H5_TUNE2
430

RV14
+12V
H5_TUNE2
1 3
GND
200

R39
H5_BPF2
1 2
H5_INV2
470k

C18
H5_NODE2
1 2
H5_BPF2
2n

C19
H5_NODE2
1 2
H5_INV2
2n

U10
+12V
GND
3 +A
H5_INV1
2 −A
OUT 1
H5_BPF1
GND
5 +B
H5_INV2
6 −B
OUT 7
H5_BPF2
−12V
TL082CP

R40
H5_BPF2
1 2
H5_APINV1
10k

R41
H5_AP1
1 2
H5_APINV1
10k

R42
H5_BPF2
1 2
H5_PHR1
100

RV15
+12V
H5_PHR1
1 3
H5_LP1
200k

C20
H5_LP1
1 2
GND
2n

R43
H5_AP1
1 2
H5_APINV2
10k

R44
H5_AP2
1 2
H5_APINV2
10k

R45
H5_AP1
1 2
H5_PHR2
100

RV16
+12V
H5_PHR2
1 3
H5_LP2
200k

C21
H5_LP2
1 2
GND
2n

U11
+12V
H5_LP1
3 +A
H5_APINV1
2 −A
OUT 1
H5_AP1
H5_LP2
5 +B
H5_APINV2
6 −B
OUT 7
H5_AP2
−12V
TL082CP

R46
H5_AP2
1 2
H5_POLINV
10k

R47
H5_NEG
1 2
H5_POLINV
10k

J3
H5_AP2
H5_SELECT
H5_NEG
1
2
3
PHASE_POLARITY

RV17
+12V
H5_SELECT
1 3
GND
10k

U12
+12V
GND
3 +A
H5_POLINV
2 −A
OUT 1
H5_NEG
H5_AMPLITUDE
5 +B
H5_OUT
6 −B
OUT 7
H5_OUT
−12V
TL082CP

TP10
H5_BPF2

TP11
H5_AP2

TP12
H5_OUT

NE555 + TL082 / +/−12 V
Sheet: /harmonic_5/
File: harmonic_5.kicad_sch
Title: Waveform synthesis / harmonic_5
Size: A2
Date: 2026-10-29
Rev: A
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All connections use named global nets. VCC = +12V; VEE = −12V; GND = 0V. See design notes for topology and calibration.

R48
SQ
1 2
H7_NODE1
220k

R49
H7_NODE1
1 2
H7_TUNE1
430

RV18
H7_TUNE1
1 3
GND
200

R50
H7_BPF1
1 2
H7_INV1
470k

C22
H7_NODE1
1 2
H7_BPF1
1.5n

C23
H7_NODE1
1 2
H7_INV1
1.5n

R51
H7_BPF1
1 2
H7_NODE2
220k

R52
H7_NODE2
1 2
H7_TUNE2
430

RV19
H7_TUNE2
1 3
GND
200

R53
H7_BPF2
1 2
H7_INV2
470k

C24
H7_NODE2
1 2
H7_BPF2
1.5n

C25
H7_NODE2
1 2
H7_INV2
1.5n

U13
VCC
GND
+A
−A
OUT
H7_BPF1
GND
H7_INV1
+B
−B
OUT
H7_BPF2
H7_INV2
VEE
TL082CP

R54
H7_BPF2
1 2
H7_APINV1
10k

R55
H7_AP1
1 2
H7_APINV1
10k

R56
H7_BPF2
1 2
H7_PHR1
100

RV20
H7_PHR1
1 3
H7_LP1
200k

C26
H7_LP1
1 2
GND
1.5n

R57
H7_AP1
1 2
H7_APINV2
10k

R58
H7_AP2
1 2
H7_APINV2
10k

R59
H7_AP1
1 2
H7_PHR2
100

RV21
H7_PHR2
1 3
H7_LP2
200k

C27
H7_LP2
1 2
GND
1.5n

U14
VCC
H7_LP1
+A
H7_APINV1
−A
OUT
H7_AP1
H7_LP2
+B
H7_APINV2
−B
OUT
H7_AP2
VEE
TL082CP

R60
H7_AP2
1 2
H7_POLINV
10k

R61
H7_NEG
1 2
H7_POLINV
10k

J4
H7_AP2
H7_SELECT
H7_NEG
1
2
3
PHASE_POLARITY

RV22
H7_SELECT
1 3
GND
10k

U15
VCC
GND
+A
H7_POLINV
−A
OUT
H7_NEG
H7_AMPLITUDE
+B
H7_OUT
−B
OUT
H7_OUT
VEE
TL082CP

TP13
H7_BPF2

TP14
H7_AP2

TP15
H7_OUT

NE555 + TL082 / +/−12 V
Sheet: /harmonic_7/
File: harmonic_7.kicad_sch
Title: Waveform synthesis / harmonic_7
Size: A2
Date: 2026-10-29
Rev: A
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All connections use named global nets. VCC = +12 V; VEE = −12 V; GND = 0 V. See design notes for topology and calibration.

R62
H1_OUT
1
2
SUM_INV
100k

R63
H3_OUT
1
2
SUM_INV
100k

R64
H5_OUT
1
2
SUM_INV
100k

R65
H7_OUT
1
2
SUM_INV
100k

R66
SUM_NEG
1
2
SUM_INV
100k

R67
SUM_NEG
1
2
RESTORE_INV
100k

R68
SUM_RAW
1
2
RESTORE_INV
100k

U16
VCC
GND
+A
SUM_INV
-A
GND
+B
RESTORE_INV
-B
OUT
SUM_NEG
OUT
SUM_RAW
VEE
TL082CP

R69
SUM_RAW
1
2
OUT
100

TP16
SUM_RAW
SUM_RAW

TP17
OUT
OUT

TP18
GND
GND

J5
VCC
GND
VEE
1
2
3
POWER_12V

J6
OUT
GND
1
2
WAVE_OUT

C28
VCC
1
2
GND
47u/25V

C29
GND
1
2
VEE
47u/25V

C30
VCC
1
2
GND
100n

C31
VCC
1
2
GND
100n

C32
VCC
1
2
GND
100n

C33
GND
1
2
VEE
100n

C34
VCC
1
2
GND
100n

C35
GND
1
2
VEE
100n

C36
VCC
1
2
GND
100n

C37
GND
1
2
VEE
100n

C38
VCC
1
2
GND
100n

C39
GND
1
2
VEE
100n

C40
VCC
1
2
GND
100n

C41
GND
1
2
VEE
100n

C42
VCC
1
2
GND
100n

C43
GND
1
2
VEE
100n

C44
VCC
1
2
GND
100n

C45
GND
1
2
VEE
100n

C46
VCC
1
2
GND
100n

C47
GND
1
2
VEE
100n

C48
VCC
1
2
GND
100n

C49
GND
1
2
VEE
100n

C50
VCC
1
2
GND
100n

C51
GND
1
2
VEE
100n

C52
VCC
1
2
GND
100n

C53
GND
1
2
VEE
100n

C54
VCC
1
2
GND
100n

C55
GND
1
2
VEE
100n

C56
VCC
1
2
GND
100n

C57
GND
1
2
VEE
100n

C58
VCC
1
2
GND
100n

C59
GND
1
2
VEE
100n

#FLG1
VCC
POWER
PWR_FLAG

#FLG2
GND
POWER
PWR_FLAG

#FLG3
VEE
POWER
PWR_FLAG

**NE555 + TL082 / +/−12 V**

Sheet: /power_sum/
File: power_sum.kicad_sch

***Title: Waveform synthesis / power_sum***

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-29 | **Rev: A** |
| KiCad E.D.A. 9.0.2 | | Id: 7/7 |